

BILAL AHAMAD

Lead Embedded Firmware & Systems QA Engineer

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SUMMARY

Quality engineering leader with 18+ years validating firmware and embedded systems for Amazon (Alexa devices), Google (VR), Rivian, Cruise, and Samsara. Builds the test architectures behind safety-critical and AI/ML-driven products — hardware-in-the-loop rigs, CI/CD device labs, and automation frameworks that have delivered \$3M+ in cost savings and \$2.1M in new revenue. Currently architecting 0-to-1 validation for a sensor-integration IDE and AI-augmented test pipelines.

CORE SKILLS

Test & Automation: Python, test framework architecture, hardware-in-the-loop (HIL), CI/CD (Jenkins), Docker/QEMU, Appium, Selenium, ADB & Android, QNX/C++

Domains: Firmware & IoT validation, automotive infotainment & V2X, AI/ML system testing, video quality, Bluetooth/Wi-Fi (WFA) certification, OTA updates

AI-Augmented Engineering: Claude Code, Cursor, Gemini — AI pair-programmed test frameworks, agentic automation pipelines

Software & Data: TypeScript, Node.js, React/Next.js, Pandas, Plotly & Splunk dashboards, AWS, GitHub Actions

EXPERIENCE

Founding System Architect & Technical QA Lead — [Stealth Mode](#) Sep 2025 - Present · Remote / Stealth

Leading architecture and validation for a Unified Sensor Integration IDE from concept to MVP, automating OEM dependency integration and standardizing sensor-level code extensions.

- Engineered a pre-validated compatibility layer for industry-standard chips (ARM, ESP32) and sensors (LiDAR/Radar), cutting manual bring-up cycles by an estimated 70%.
- Built a containerized Docker/QEMU test bench that runs A/B validation and system-level performance engineering before any physical hardware exists.

Senior Firmware Quality Lead — [Samsara Inc](#) Dec 2023 - Jul 2025 · Remote / San Francisco, CA

Led end-to-end system test strategy for the IoT Dash Cam product series within an AI/ML video-based safety ecosystem.

- Built automated test pipelines for video quality, AI/ML algorithm accuracy, and workload performance across the Dash Cam product series.
- Cut regression cycles by 50% and accelerated release cadence by automating network reliability and connectivity testing.

Senior Automation Engineer II — [Cruise LLC](#) Oct 2022 - Jun 2023 · San Francisco, CA

Owned test delivery for vehicle computational systems, taking safety-critical firmware versions through release.

- Developed an automation test framework for Sanity, Integration, and Stress testing, achieving 75% test coverage.
- Led design for fault tolerance, redundancy, and network reliability testing (V2X and PCAP replay).

Infotainment & V2X Firmware Test Lead — Rivian Automotive LLC Jun 2021 - Sep 2022 · Palo Alto, CA
Directed technical testing strategy and E2E test delivery for in-car Infotainment Systems in R1T, R1S, and Fleet models.

- Designed an automation test framework for QNX and Android platforms covering OTA updates, UI/UX, and navigation.
- Ran integration, performance, and stability testing across all three vehicle programs (R1T, R1S, Fleet) through continuous release cycles.

Senior QA Engineer II & Test Lead — Amazon Lab126 Jun 2018 - Jun 2021 · Sunnyvale, CA
Managed the QA process for Alexa Voice Service (AVS)-enabled products including Echo Buds (1st & 2nd Gen), Echo Auto, and Dash Button.

- Automated Snippet API library, reducing manual testing costs by \$3M.
- Set up test labs and CI/CD pipelines running continuously across 50+ devices, achieving 30% fewer post-launch issues.

Senior Test Engineer — Tech Mahindra / Google Inc Jan 2016 - Jun 2018 · Mountain View, CA
Led system testing for Google's VR Controller program, shipping 1st & 2nd Gen controllers and the Daydream View headset.

- Engineered a 3DOF robotic arm using Arduino and servo motors for automated motion tracking testing, reducing execution hours by 80%.
- Identified VR test features and built coverage across all three shipping products, closing gaps in product readiness.

Product Validation Engineer — Cognizant Technology / Cisco Sep 2015 - Jan 2016 · Boston, MA
Expanded test scope by 40% and cut execution time by 50% through modular architecture and programmable actions.

Radio Validation Engineer — Wistron Mobile Solutions Corp Dec 2014 - Sep 2015 · Rolling Meadows, IL
Re-architected the WLAN IOT test automation framework to add Android OS support, saving \$1.3M.

Senior Software Engineer & Test Automation Lead — L&T Infotech / Motorola Mobility Oct 2009 - Dec 2014 · Chicago, IL
Devised a 3-tier infrastructure enhancing code reusability (3x) and reducing test preparation LOE by 70%.

Software Developer — Luminous Infoways Oct 2008 - Sep 2009 · Bhubaneswar, India
Delivered web application deployment modules and feature integration, and served as the client liaison on portal releases.

SELECTED PROJECTS

US Live Layoff Monitoring Dashboard — A national WARN layoff dataset unifying notices from 46 states and DC into one searchable feed with a free JSON API and per-state email alerts — refreshed twice daily by a GitHub Actions pipeline with ETag caching and MD5 verification, zero human intervention.

Adhan Caster — Cross-Browser Extension — A Manifest V3 browser extension that auto-pauses every playing video and audio across all open tabs at Adhan time, with a live countdown to the next prayer. One codebase live on the Chrome Web Store, Firefox Add-ons and Edge Add-ons — built in one day with Claude Code.

Smart-Home IoT Media Caster — An IoT orchestration layer that casts automated prayer-time notifications to a Sony Android TV, built in 4 days instead of a 3-week engineering cycle with AI pair programming. Integrates a Raspberry Pi over ADB, managing media state and low-level device commands.

CERTIFICATIONS

- Google AI Professional (2026)
- Google Project Management Professional (2026)
- ISTQB Certified Tester Foundation Level (CTFL)
- Software Testing Foundations: Integrating AI into Quality Process (2026)
- AI Coding Agents with GitHub Copilot and Cursor (2025)

EDUCATION

B.Tech, Electronics & Telecommunications Engineering
Biju Patnaik University of Technology — 2004 – 2008